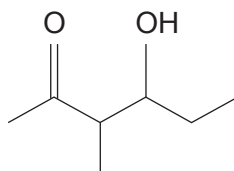


SECTION AAnswer **all** questions.

1. Give the molecular formula of the compound shown. [1]



Molecular formula

2. When fuels are burned with insufficient oxygen, incomplete combustion occurs and carbon monoxide is formed instead of carbon dioxide.

Write the equation for the incomplete combustion of propane, C_3H_8 . Assume that carbon monoxide is the only carbon-containing product. [1]

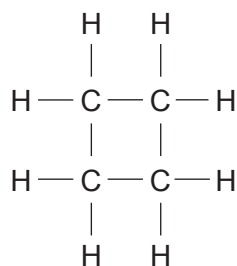
.....

3. Name the reagent that is needed to change unsaturated oils into saturated fats. [1]

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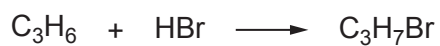
4. Cyclobutane is a cyclic hydrocarbon whose structure is shown.



Draw the structures of **two** structural isomers of this compound.

[2]

5. Propene reacts with hydrogen bromide.



Draw the structures of the **two** isomers formed and give the name of the major product.

[2]

Major product

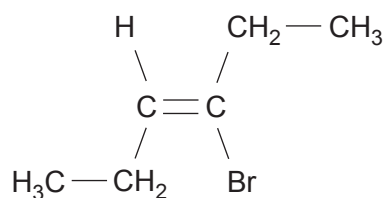


6. In the forward direction, the enthalpy change of reaction, ΔH , for a reversible endothermic reaction, has the numerical value of 60 kJ mol^{-1} .

If the activation energy, E_a , of the forward reaction is 100 kJ mol^{-1} , calculate the activation energy of the backward reaction. [1]

$E_a = \dots\dots\dots \text{ kJ mol}^{-1}$

7. Name the following *E-Z* isomer. [2]



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